

Data Quality, Structure, and Best-Use Recommendation

A Modality-Agnostic Mathematical and Algorithmic Framework

Internal Technical Specification

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Abstract

We define a unified, modality-agnostic framework to assess dataset quality, structural integrity, and task suitability across text, tabular, image, and audio. The framework combines statistical diagnostics, information-theoretic measures, white-noise tests, shift/leakage checks, and modality-specific probes to compute a multi-axis quality score and to recommend “best-use” tags. Algorithms are designed with two-pass evaluation (fast $\rightarrow$ deep), scalable deduplication, and calibrated meta-features for downstream recommendation.

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1 Notation and Data Model

Let a dataset be $D = \{(x_i, y_i, a_i)\}_{i=1}^N$ where $x_i \in \mathcal{X}$ is an item (text/tabular/image/audio), y_i an optional label, and a_i auxiliary attributes (timestamps, annotator IDs, demographics). Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ be a modality-appropriate embedding (e.g., small LM for text, CLIP for images, wav2vec for audio, numeric features for tabular), with $z_i = \phi(x_i)$.

We emit a six-to-seven axis *Quality Card* $\mathbf{S}(D) \in [0, 100]^m$ and an overall score

$$Q(D) = \sum_{j=1}^m w_j \hat{S}_j, \quad \sum_j w_j = 1, \quad \hat{S}_j \in [0, 1], \quad (1)$$

where $\hat{S}_j$ are normalized sub-scores. Typical weights: provenance 0.25, legal 0.20, statistical 0.20, annotation 0.15, technical 0.10, risk 0.10; optional information-theoretic axis, see §3.

2 Quality Scoring Axes

2.1 Axis A: Provenance and Consent (S_{prov})

Binary/ordinal checks aggregated with weights α_k :

- Creator identity verified, chain-of-custody documented, content credentials (cryptographic claims).
- Third-party licenses/releases present where applicable.

$$S_{\text{prov}} = 100 \cdot \frac{\sum_k \alpha_k b_k}{\sum_k \alpha_k}, \quad b_k \in \{0, 1, 2\}. \quad (2)$$

2.2 Axis B: Legal Clarity (S_{legal})

Rubric covering scope (train/eval/derivatives/redistribution), territory, revocation, and third-party rights. Map ordinal indicators $\{c_j\}$ to $[0, 100]$, applying fixed penalties for ambiguity or missing clauses.

2.3 Axis C: Statistical Health (S_{stats})

Label balance. For K classes with proportions $\mathbf{p}$,

$$\text{NE} = \frac{-\sum_{k=1}^K p_k \log p_k}{\log K} \in [0, 1]. \quad (3)$$

Missingness. Per-feature missing fractions m_j , aggregate $\bar{m} = \frac{1}{d} \sum_{j=1}^d m_j$.

Duplicates/near-duplicates. Estimate via MinHash/SimHash (text/tabular) or perceptual hashing (images/audio):

$$\text{dup} \approx \frac{\#\{(i, j) : \text{sim}(x_i, x_j) \geq \tau\}}{\binom{N}{2}}. \quad (4)$$

Redundancy-adjusted effective size (RAES). Build k -NN over $\{z_i\}$; with mean edge similarity ρ ,

$$\text{RAES} = \frac{N}{1 + \lambda \rho}, \quad \widehat{\text{RAES}} = \min\left(1, \frac{\text{RAES}}{N}\right). \quad (5)$$

Coverage/diversity. K -means on $\{z_i\}$ yields cluster occupancies u_c :

$$\text{cov} = 1 - \frac{1}{2} \sum_{c=1}^K \left| \frac{u_c}{N} - \frac{1}{K} \right| \in [0, 1]. \quad (6)$$

Label noise proxy. Train a small classifier h ; let ϵ_{cv} be cross-validated error.

Aggregation.

$$S_{\text{stats}} = 100 \cdot (\beta_1 \text{NE} + \beta_2(1 - \overline{m}) + \beta_3(1 - \text{dup}) + \beta_4 \widehat{\text{RAES}} + \beta_5 \text{cov} + \beta_6(1 - \epsilon_{cv})). \quad (7)$$

2.4 Axis D: Annotation Quality (S_{ann})

Agreement. For two raters: Cohen’s $\kappa = \frac{p_o - p_e}{1 - p_e}$; for multiple raters/levels: Krippendorff’s $\alpha = 1 - \frac{D_o}{D_e}$ with observed/expected disagreement D_o, D_e .

Guidelines/audits. Score presence/clarity of instructions and gold-questions accuracy.

$$S_{\text{ann}} = 100 \cdot (\gamma_1 \hat{\kappa} + \gamma_2 \hat{\alpha} + \gamma_3 \text{guidelines} + \gamma_4 \text{gold}). \quad (8)$$

2.5 Axis E: Technical Integrity (S_{tech})

Schema conformance, valid formats, checksums, versioning, deterministic splits:

$$S_{\text{tech}} = 100 - \sum_r \delta_r \cdot \mathbf{1}\{\text{rule } r \text{ failed}\}. \quad (9)$$

2.6 Axis F: Harm/PII Risk (S_{risk})

Let (P, R) be precision/recall of risk detectors on calibrated samples with prevalence π :

$$\text{FPR} = (1 - \pi)(1 - P), \quad \text{FNR} = \pi(1 - R), \quad \text{risk} = \theta_1 \text{FPR} + \theta_2 \text{FNR} + \theta_3 \cdot \text{severity}. \quad (10)$$

Then $S_{\text{risk}} = 100 \cdot (1 - \widehat{\text{risk}})$ after normalization.

3 Information-Theoretic Diagnostics (Axis G)

Shannon entropy. Discrete $H(X) = -\sum_x p(x) \log p(x)$ (normalize by $\log |\mathcal{A}|$). For continuous features use binned or differential-entropy estimators.

Entropy rate (sequences). $h = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1^n)$; use LZ-based or PPM estimators $\hat{h}$.

Mutual information. $I(X; Y) = H(X) + H(Y) - H(X, Y)$; leakage via $I(X; \text{split})$ and $I(Y; \text{ID/timestamp})$. Estimation: k NN (KSG), Miller–Madow bias-corrected discrete.

Total correlation. $TC(X) = \sum_i H(X_i) - H(X)$ (feature redundancy).

Distributional divergence. $\text{KL}(P\|Q)$ or $\text{JS}(P\|Q)$ vs. domain references.

Compression signals. Normalized Compression Distance (NCD) and chunk compressibility (e.g., gzip/brotli) as MDL proxies.

Perplexity delta (text). $\Delta \text{PPL} = (\text{PPL}_{\text{gen}} - \text{PPL}_{\text{dom}}) / \text{PPL}_{\text{gen}}$.

Info axis score (example).

$$S_{\text{info}} = 100 \left[0.25 \widehat{I(X; Y)} + 0.20 (1 - \widehat{TC}) + 0.15 \widehat{\Delta \text{PPL}} / \text{JS} + 0.20 (1 - |\hat{h} - h_{\text{ref}}|) + 0.20 (1 - \text{compressibility-z}) \right]. \quad (11)$$

Flag if $I(X; \text{split})$ is high or $\hat{h} \approx$ white-noise levels.

4 Structure, Shift, and Leakage Diagnostics

Let A, B be two samples (e.g., train/test).

Maximum Mean Discrepancy (MMD).

$$\text{MMD}^2 = \left\| \frac{1}{|A|} \sum_{x \in A} \psi(x) - \frac{1}{|B|} \sum_{y \in B} \psi(y) \right\|_{\mathcal{H}}^2, \quad \psi \text{ induced by kernel } k. \quad (12)$$

Energy distance.

$$\mathcal{E}^2(A, B) = 2 \mathbb{E} \|X - Y\| - \mathbb{E} \|X - X'\| - \mathbb{E} \|Y - Y'\|. \quad (13)$$

Wasserstein-1 (1D). For equal-size samples,

$$W_1 = \frac{1}{n} \sum_{i=1}^n |X_{(i)} - Y_{(i)}|. \quad (14)$$

Classifier Two-Sample Test (C2ST). Train a small classifier to distinguish A vs B ; $\text{AUC} \gg 0.5$ indicates shift.

Leakage tests. Train h to predict split s from (x, y, a) ; high AUC or high $I(s; x, y, a)$ indicates split artifacts. Target leakage via $I(y; \text{IDs}/\text{timestamps})$.

5 White-Noise / Randomness Tests (Residual-First)

Fit a simple baseline, compute residuals r_t , then test whiteness.

Ljung–Box Q -statistic.

$$Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}, \quad \hat{\rho}_k \text{ autocorrelation at lag } k. \quad (15)$$

BDS test (i.i.d.). For embedding dimension m and radius ϵ , compare correlation integrals $C_m(\epsilon)$ to $C_1(\epsilon)^m$; large standardized deviation rejects i.i.d.

Runs test. Wald–Wolfowitz runs against the median for independence.

Spectral flatness. With PSD bins $\{P_k\}$,

$$\text{SFM} = \frac{\exp\left(\frac{1}{K} \sum_{k=1}^K \ln P_k\right)}{\frac{1}{K} \sum_{k=1}^K P_k} \in (0, 1], \quad (16)$$

near 1 for white noise.

Permutation entropy. $H_\pi(m, \tau)$ normalized to $[0, 1]$; high values $\approx$ random.

Modality tips. Text: whiten surprisal residuals of n -gram LM; Images: test 2D-PSD flatness on high-frequency residuals; Audio: ARMA residual SFM/ACF; Tabular time series: Ljung–Box on per-feature residuals and model residuals.

6 Modality-Specific Probes

Text. Δ PPL, lexical diversity, embedding isotropy/effective rank, toxicity/PII prevalence, dedup at scale (LSH).

Images. Variance of Laplacian (sharpness), CLIP diversity and linear probe accuracy, SSIM vs references, perceptual hashes (pHash/dHash) for dedup.

Audio. SNR (segmental), baseline WER, speaker diversity, VAD-based silence ratios, zero-crossing rate, ACF.

Tabular. Mutual information (or Predictive Power Score), monotonicity checks, linear vs tree baseline gap, time/ID leakage diagnostics.

7 Best-Use Recommendation Engine

7.1 Meta-Features

Construct $\varphi(D)$ from core metrics: NE, $\overline{m}$, dup, $\widehat{\text{RAES}}$, cov, ϵ_{cv} , MMD/energy, I -based leakages, S_{info} components ($\hat{h}$, compressibility z , TC , Δ PPL, WER, sharpness, CLIP diversity).

7.2 Rule-Based Mapping (transparent)

Classification (labeled) $\Leftarrow$ NE $\in [0.6, 0.95]$, $\kappa > 0.7$, dup < 0.05 , $\widehat{\text{RAES}} > 0.6$.

Text Pretraining/RAG $\Leftarrow$ Δ PPL > 0.3 , tox/PII low, cov > 0.7 .

ASR Fine-tune $\Leftarrow$ $20\% < \text{WER} < 50\%$, SNR > 15 dB, speaker diversity high.

Vision Pretrain $\Leftarrow$ sharpness moderate, CLIP diversity high, dup low.

7.3 Meta-Model (learned, calibrated)

Train a small classifier $g : \varphi(D) \mapsto \{\text{tags}\}$ with calibrated probabilities (Platt/Isotonic). Provide top- k tags and SHAP-style explanations.

8 Two-Pass Evaluation Pipeline

Algorithm 1: Assess-and-Recommend Pipeline

Input: Dataset D , modality, label presence flag

Output: Scores $\mathbf{S}(D)$, overall $Q(D)$, tags, diagnostics, Quality Card JSON

Pass 1 (sampled, fast):

Schema/format checks; checksums; versioning.

Sample $S \subset D$; compute embeddings $z = \phi(S)$.

Stats: NE, $\overline{m}$, LSH-dup, $\widehat{\text{RAES}}$ (approx), cov, linear-probe CV.

Risk scan: PII/toxicity (precision/recall-calibrated).

Info: H , $\hat{h}$, compressibility, NCD-samples, I leak screens.

Pass 2 (deep, conditional):

Full dedupe (ANN/graph), shift tests (MMD/energy/C2ST), annotation κ/α , modality probes (CLIP/WER/ Δ PPL, sharpness).

White-noise tests on residuals (Ljung-Box, BDS, SFM, permutation entropy).

Aggregate:

Compute axes $S_{\text{prov}}, S_{\text{legal}}, S_{\text{stats}}, S_{\text{ann}}, S_{\text{tech}}, S_{\text{risk}}, S_{\text{info}}$.

$Q(D) = \sum_j w_j \hat{S}_j$; map to grade {A,B,C,D}.

Best-use: rule-based $\cup g(\varphi(D))$ with calibrated confidence.

Emit Quality Card JSON + audit fingerprints.

Pythonic pseudocode (sketch).

```
def assess_dataset(D, modality, has_labels):
    meta = run_technical_checks(D)
    S = sample_for_fast_pass(D, n=min(10000, len(D)))
    z = embed(S, modality) # text: small LM; image: CLIP; audio:
    wav2vec; tab: numeric
    stats = compute_stats(S, z) # NE, missingness, dup(LSH),
    RAES, cov, linear-probe
    risk = risk_scan(S, modality) # PII/toxicity; prevalence-
    calibrated
    info = info_signals(S, z, labels=get_labels(S)) # H, h, MI, TC,
    compressibility, NCD

    deep = {}
    if needs_deep_pass(stats, risk, info, size=len(D)):
        deep["dedupe"] = full_dedupe(D, modality) # ANN/graph-based
        deep["shift"] = shift_tests(D, z) # MMD, energy,
        C2ST
        deep["annq"] = annotation_quality(D) # kappa/alpha +
        gold audits
        deep["probes"] = modality_probes(D, modality) # PPL, WER,
        sharpness...
        deep["white"] = whiteness_on_residuals(D, modality)

    scores = aggregate_axes(meta, stats, risk, info, deep)
    tags = best_use(scores, stats, info, deep)
    card = build_quality_card(D, scores, tags, diagnostics={**stats,
        **risk, **info, **deep})
    return scores, tags, card
```

9 Complexity and Scaling

- **Deduplication:** Use LSH/ANN (HNSW/IVF-PQ) for sublinear near-duplicate search; full pairwise only on flagged subsets.
- **Graphs:** k -NN graph with small k (10–30); compute RAES and coverage on sampled nodes for very large N .
- **Shift tests:** Run on embeddings with capped sample size (e.g., $n \leq 50k$); use block bootstrap for CIs.
- **Info measures:** Reduce dimensionality (PCA/random projections) before MI/TC; use shuffled baselines for bias control.

10 Default Thresholds and Grading

Metric	Good (default)	Flag if
duplication rate	< 5%	> 10%
missingness $\overline{m}$	< 5%	> 15%
$\widehat{\text{RAES}}$	> 0.6	< 0.4
coverage (cov)	> 0.7	< 0.5
Cohen κ	> 0.7	< 0.5
MMD/C2ST (AUC)	≤ 0.6	≥ 0.7
$I(X; \text{split})$	$\approx$ shuffled	> 2σ over shuffle
SFM residuals	< 0.8	> 0.9 (too white)
Permutation entropy (resid.)	< 0.85	> 0.9

Grade mapping (default): A: [90, 100], B: [80, 90), C: [65, 80), D: < 65.

11 Outputs: Quality Card JSON (Schema Sketch)

```
{
  "dataset_id": "uuid",
  "modality": "text|image|audio|tabular",
  "size": N,
  "scores": {
    "provenance": 92, "legal": 88, "stats": 84,
    "annotation": 81, "technical": 95, "risk": 86, "info": 89,
    "overall": 88, "grade": "B"
  },
  "diagnostics": {
    "NE": 0.82, "missing_mean": 0.03, "dup_rate": 0.04,
    "RAES_hat": 0.71, "coverage": 0.76, "cv_error": 0.22,
    "MMD": 0.04, "C2ST_AUC": 0.58, "MI_split": 0.002,
    "entropy": 4.21, "entropy_rate": 3.12,
    "TC_hat": 0.37, "compressibility_z": 0.6,
    "white_raw": false, "white_resid": false, "periodic_artifact":
      false,
    "text": {"delta_ppl": 0.36}, "image": {"sharpness": 0.48, "clip_div
      ": 0.71},
    "audio": {"snr_db": 19.4, "wer": 0.34}, "tabular": {"pps_max":
      0.41}
  },
  "best_use": [
    {"tag": "classification", "confidence": 0.84},
    {"tag": "text-pretraining", "confidence": 0.77}
  ]
}
```

```
],
"provenance": {"creator_verified": true, "credentials": true},
"legal": {"license": "train-only-nonexclusive", "territory": ["EU", "US", "UK", "CA"]},
"audit": {"hashes": ["..."], "timestamp": "2025-10-16T12:34:56Z"}
}
```

12 Implementation Notes

Python stack: `numpy/scipy/pandas`, `scikit-learn` (tests, MI approximations, calibration), `faiss/hnswlib` (ANN), `imagehash/opencv` (perceptual dedup), `librosa/jiwer` (audio SNR/WER), `cleanlab` (label noise), `presidio` (PII), light `transformers` models for embeddings/perplexity, `great-expectations/evidently` for data checks. Use Arrow/Parquet; cache embeddings; persist audit logs (hash chains or Merkle DAG).

Disclaimer. Thresholds/weights herein are sane defaults and should be tuned per domain. Statistical tests assume sufficient sample sizes and appropriate independence conditions.